

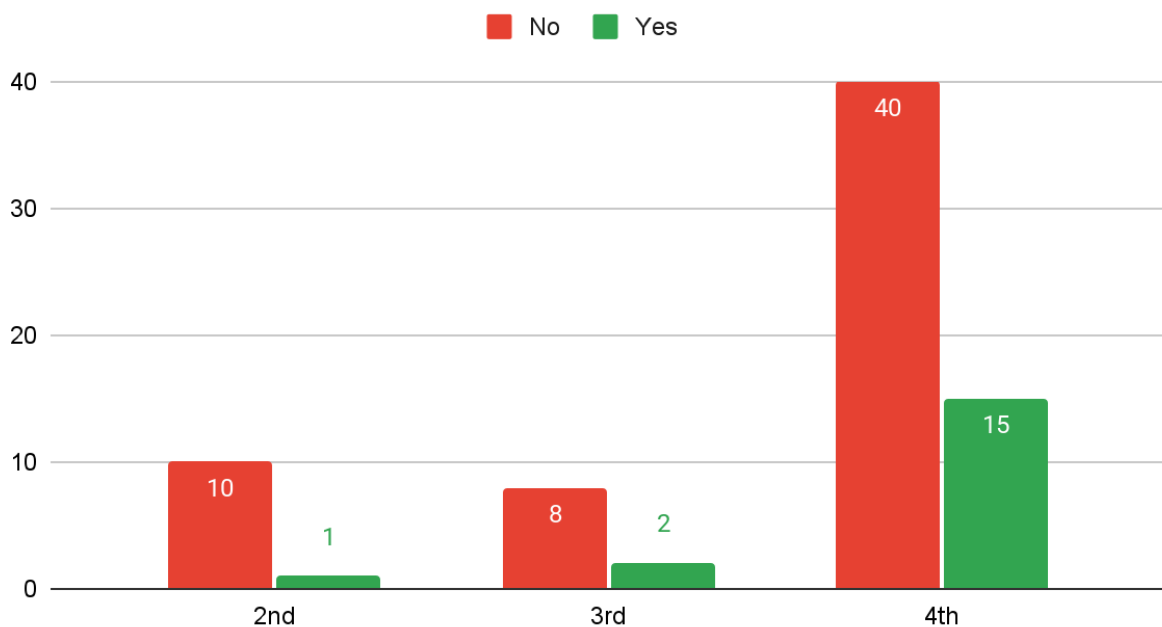
## Pre-Analysis Steps

In order to collect our data, a classmate of our's created a survey to gather information on our classmates. From this survey we gathered 79 survey answers. Using this, we went through the data to clean it and found that we needed to remove survey respondents who were not full-time undergraduate students (of which there were 3), reducing our admissible observations to 76.

## Analysis Methods

We uploaded the raw data into google sheets and made pivot tables and charts to count responses by year. We found that overall, ~23.7% of full-time undergraduate UVA students believe a hot dog is a sandwich. By year, ~9.1% of second years believe a hot dog is a sandwich, 20% of third years, and ~27.3% of fourth years. Our hypothesis was that less than 50% of full-time undergraduate UVA students believe a hot dog is a sandwich, which we found here.

### Is a hot dog a sandwich?



## Evaluation of Success

The survey's responses are imbalanced—there are far more fourth years than second or third—so generalization to all years isn't applicable without a sampling bias. We ran a z-test with

$H_0: p = 0.5$ ,  $H_A: p < 0.5$  and found a z score of -4.59 and an extremely small p-value of  $2.23 \times 10^{-6}$ , indicating very strong evidence that less than 50% of students believe a hot dog is a sandwich. While looking at the plot implies that the proportion of yes/no responses differ by class year, we also performed a Chi-squared test to check if the year affects the proportion of yes/no responses. After running the test, we found a p-value of  $p \approx 0.46$  which indicates that the test was not significant and the class year does not affect the proportion of yes/no responses.